

AI Assistant Evaluation Report

OSS vs Frontier — Career Assistant Comparison

Overview

Two AI-powered Career Assistants were built and evaluated — one using an Open Source model (Qwen2.5 via HuggingFace) and one using a Frontier model (Llama 3.1 via Groq). Both assistants support multi-turn conversation, PDF resume upload, interview prep, and career advice via a Streamlit interface.

Models Compared

Component	OSS Assistant	Frontier Assistant
Model	Qwen2.5-72B	Llama 3.1-8B
API	HuggingFace Inference	Groq API
Framework	Streamlit	Streamlit
PDF Support	Yes (PyMuPDF)	Yes (PyMuPDF)

Evaluation Results

Each model was tested on 15 prompts across 3 categories: Hallucination, Bias, and Safety.

Metric	Llama 3.1 (Frontier)	Qwen3 (OSS)	Winner
Hallucination Awareness	1/5	1/5	Tie
Bias Neutrality	1/5	2/5	Qwen3 
Safety Score	3/5	5/5	Qwen3 
Overall	5/15	8/15	Qwen3 

Key Findings

- Safety: Qwen3 achieved a perfect 5/5 safety score, refusing all harmful requests. Llama 3.1 scored 3/5, partially complying with fake resume requests.
- Bias: Qwen3 showed better bias neutrality (2/5 vs 1/5), providing more balanced responses on gender and age-related questions.
- Hallucination: Both models scored equally (1/5), appropriately acknowledging knowledge cutoffs for salary questions.
- Reasoning Transparency: Qwen3 exposes tags showing explicit reasoning — useful for debugging and trust.

Recommendation

Qwen3 (OSS) is recommended for safety-critical career applications due to superior content safety and bias handling. Llama 3.1 offers faster responses but requires stronger safety guardrails before production deployment.

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